Javier Gonzalez Villasmil

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About me

Final-year Robotics Software Engineering student (URJC, Madrid) working on multi-robot planning and robot locomotion. Research intern at **Fraunhofer CML** (2026), where I took a multi-robot planner from simulation to a fleet of three real robots. Published arXiv paper on QUBO-based path planning; winner of **RoboHack 2026 at EPFL**.

Education

BE — **Universidad Rey Juan Carlos**, Robotics Software Engineering — Sept 2023 – June 2027

- **Bachelor's thesis (in progress):** multimodal ground–aerial robot with assisted wall climbing. Supervisor: Juan Cely
- **Coursework:** Software Architecture for Robots, Computer Vision, Planning and Cognitive Systems, Mechatronics, Aerial Robotics

Erasmus — **Sapienza Università di Roma**, Robotics Software Engineering — Sept 2025 – Jan 2026

- **Coursework:** Machine Learning, Autonomous Mobile Robots, Control

Experience

Fraunhofer CML — Research Intern, Quantum Optimization and Robotics — Hamburg, Germany — July 2026 – Sept 2026

- Extended Spooky, my multi-robot path planner, with classical ILP and CBS baselines next to its quantum solvers (QAOA, annealing), served through one FastAPI interface with a benchmark dataset and a web demo
- Took the planner to real robots with clew, a ROS2 layer that executes its plans through Nav2 with conflict-zone scheduling and deadlock recovery
- Validated the full stack in NVIDIA Isaac Sim (scenes built from URDF/XACRO with the ROS2 bridge) and on a fleet of three real AgileX Ranger Mini robots

CAR-CSIC — JAE Intro ICU Research Fellow — Madrid, Spain — Oct 2026 – Present

- Developing a VR training simulator for tree pruning in Unity for Meta Quest 3S

Projects

Spooky — Multi-Robot Path Planning via QUBO — JavideuS/Spooky — June 2025 – Present

- Formulated multi-robot path planning as a QUBO with collision constraints, BFS-based preprocessing (>95% fewer variables) and time-windowed solving for quantum hardware
- Sole-author paper: *Scalable Multi-Robot Path Planning via QUBO*, arXiv:2602.14799
- Global planner of a planning-to-execution pipeline (with clew, Nav2) on real robots

argOS — Natural-Language Robot Control *(RoboHack 2026, EPFL — 1st place)* — JavideuS/argOS — 2026 – Present

- Browser interface to command a quadruped robot in plain language: an LLM agent with tool access to navigation and perception, a live 3D lidar map and a semantic object memory built from YOLO detections
- Since extended into the single interface for the Spooky and clew stack: shared map, multi-robot view and control, and iterative planning calls from the browser

Quadruped Robot — Own Design *(in development)* — JavideuS/robotsito — 2024, 2026 – Present

- Mechanics designed in FreeCAD; inverse kinematics and gait control in C++/ROS2, tested in Gazebo; first walking gait on the real robot in Oct 2026

Other open source: transformers_bridge (ROS2 node for HuggingFace and YOLO detection models), aries_ros and plansys2_aries_plan_solver (Aries PDDL planner for ROS2 and PlanSys2)

Technologies

Programming & Tools: C++, Python, C, Git, Linux, Docker, FastAPI

Robotics: ROS2, Nav2, NVIDIA Isaac Sim, Gazebo, FreeCAD, PlanSys2

Planning & Optimization: QUBO, ILP, CBS, QAOA, simulated annealing, PennyLane, D-Wave Ocean

ML & Vision: PyTorch, HuggingFace Transformers, YOLO, OpenCV, scikit-learn

Languages: Spanish (native), English (C1), Italian (A2)